

ACHYUTHAN RAGHAVAN

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EDUCATION

MSc Physics with BE Electronics and Communication Engineering

Aug 2023 – Present

Birla Institute of Technology and Science (BITS), Pilani

EXPERIENCE

Electronics Intern, *Ecocipher, Gruner Renewable Energy*

Jun 2026 – Present

- Built a 5 kg AUW fixed-wing drone from scratch for agricultural mapping, covering full system integration including frame assembly, motor and ESC selection, power distribution wiring, flight controller configuration, and pre-flight calibration.
- Configured and validated pre-planned autonomous missions using ArduPilot/Mission Planner, successfully executing waypoint-based survey flights over agricultural fields with stable performance.
- Performed basic image processing tasks on aerial imagery captured during mapping missions, supporting early-stage orthomosaic and field analysis workflows.
- Gained foundational understanding of Field Oriented Control (FOC) principles and assisted in debugging a previously designed custom inverter used to drive a 3-phase PMSM motor, including micro-soldering work on the board.

Electrical Subsystem Lead, *Criss Robotics*

Apr 2024 – Mar 2026

- Challenged with architecting a full-stack electrical system for a Mars-analog rover from scratch; led a team of 5+ engineers to deliver a working prototype, reducing integration issues by ~30% through modular design practices.
- Identified signal integrity bottlenecks across microcontroller-to-actuator interfaces; redesigned PCB layouts in KiCad and Altium for ESP32 and Arduino-based subsystems, cutting wiring errors by 40% and improving system reliability in field tests.
- Designed a significantly more compact PCB layout using a stackable PCB architecture, reducing external wiring complexity and resulting in a more compact and maintainable rover form factor.
- Integrated absolute and incremental encoders with actuator systems using potentiometer feedback and limit switches, enabling precise positional control and reliable end-stop detection across rover subsystems.

Project Intern, *IGCAR – Indira Gandhi Centre for Atomic Research, Kalpakkam*

May 2025 – Jul 2025

- Tasked with developing a field-deployable turbidity sensor for nuclear facility water monitoring; designed PCB from scratch using ATmega2560, integrating analog sensing circuitry with digital readout — delivering a functional prototype within an 8-week timeline.
- Faced challenge of unreliable off-the-shelf microcontroller modules; transitioned to custom ATmega2560 PCB design, reducing component costs by ~35% and improving sensor accuracy by replacing the LDR-based sensing approach with a TSL235R light-to-frequency converter, achieving significantly higher measurement precision.
- Created complete technical documentation including schematic drawings and BOM; designed and 3D-printed a ruggedized housing using Fusion 360.

President, *Renewable Energy Club, BITS Pilani*

Apr 2025 – Apr 2026

- Leading a team of 6+ members in the design and development of auto-aligning solar panels with active solar tracking to improve energy capture efficiency.
- Conducted for the first time a campus-wide Solar RC Racing competition and Cyclotron event, bringing together students across departments to engage with renewable energy applications.
- Managed Solar Cooking event that raised over INR 10,000 which was donated entirely to Nirmaan, a student-run social welfare society, demonstrating community-focused initiative.

PROJECTS

Solar Tracking Solar Panel

- Developing an auto-aligning solar panel system with a servo-driven dual-axis tracker, using light-sensor feedback to continuously orient panels toward maximum solar irradiance.

ESP32-Based Game Boy

- Built a handheld retro gaming console using an ESP32 microcontroller with a TFT display and custom PCB, with joystick and button input handling.

IoT-Based RC Cars

- Built Wi-Fi-controlled RC cars using ESP32, designing the motor driver circuitry and firmware for real-time wireless control through a custom web interface.

TECHNICAL SKILLS

PCB Design & EDA: KiCad, Altium Designer

Embedded Systems: Arduino, ESP32

Hardware & Prototyping: 3D Printing, Fusion 360, Cabling & Grounding, Sensor Integration, Soldering

Soft Skills: Public Speaking, Team Leadership, Cross-functional Collaboration

EXTRACURRICULAR ACTIVITIES & ACHIEVEMENTS

Debates & MUNs

- Attended, chaired, and won multiple Model UN conferences and competitive debate tournaments, building strong analytical and communication skills.

Mridangist

- Performed on Mridangam at classical concerts and college cultural productions, demonstrating discipline and teamwork in high-stakes performance settings.

Basketball

- Represented school basketball team in inter-school competitions, developing a collaborative and competitive mindset.